Homework 09: Support vector machines

cgao19 | October 25, 2026 | Python

Question 1. Choosing an SVM kernel

Data and protocol. The supplied SAheart data contain 462 rows, 9 predictors, and class counts (302, 160) for chd=(0,1). Predictors are sbp, tobacco, ldl, adiposity, famhist, typea, obesity, alcohol, and age; famhist is Absent=1, Present=2. There are no missing values, constant predictors, or duplicate rows. Five shuffled stratified folds (seed 43209; sizes 93,93,92,92,92) are reused for every setting. Each scaler is fitted only on that fold's fitting rows.

18 settings. Linear and degrees 2,3,4 each use C in {0.1,1,10}; radial crosses these costs with gamma in {0.01,0.1}. On standardized vectors z,w, polynomial $K=(1+z'w/9)^d$ fixes gamma=1/9 and coef0=1; radial $K=\exp(-\text{gamma}\ ||z-w||^2)$. All fits use unweighted classes, tolerance 1e-7, no probability fitting, and the zero decision-score cutoff.

ID	Kernel	C	Degree	gamma	CV error (%)	SE (pp)
1	Linear	0.1	--	--	27.5	1.9
2	Linear	1	--	--	27.7	2.6
3	Linear	10	--	--	27.7	2.6
4	Quadratic	0.1	2	1/9	26.8	0.7
5	Quadratic	1	2	1/9	27.3	0.9
6	Quadratic	10	2	1/9	28.8	0.4
7	Cubic	0.1	3	1/9	27.3	0.9
8	Cubic	1	3	1/9	30.1	1.6
9	Cubic	10	3	1/9	34.2	1.2
10	Degree 4	0.1	4	1/9	27.5	1.1
11	Degree 4	1	4	1/9	32.0	2.0
12	Degree 4	10	4	1/9	36.8	2.2
13	Radial	0.1	--	0.01	34.6	0.1
14	Radial	0.1	--	0.1	34.4	0.2
15	**Radial**	**1**	**--**	**0.01**	**26.2**	**1.1**
16	Radial	1	--	0.1	28.8	0.7
17	Radial	10	--	0.01	27.1	0.8
18	Radial	10	--	0.1	32.7	1.5

CV error is the arithmetic mean of five validation error rates; SE = sample SD / sqrt(5), in percentage points (pp).

Preference. I prefer the radial setting C=1, gamma=0.01 (ID 15) by minimum mean CV error: **26.2%** (SE 1.1 pp). The best quadratic model has error 26.8%, and the best linear model 27.5%. Thus the observed advantage over quadratic is only 0.6 pp; it does not establish that radial SVM is superior. The majority-class error is 34.6%. Increasing C hurts the higher-degree polynomials here, consistent with excessive flexibility. Very small C makes the radial fits nearly majority-only. The grid is small, and the minimum CV error is an optimistic selection estimate; fold dependence also makes the SE descriptive, rather than a confidence interval.

Question 2. Skill design, invocation, and checks

Changes motivated by both peer reviews

The review of my Homework 7 skill found an unclear definition of simplicity for LDA/QDA blends, a missing seed and classification rule, and a report that gave test errors for multiple models despite the instruction to evaluate only the selected model. The new SVM skill replaces covariance-parameter counting with a fully stated one-standard-error selection convention: linear, then polynomial degrees 2,3,4, then radial; within a family prefer smaller C, then smaller gamma, then lower CV error. This ordering expresses a preference for a simpler explanation; it is not a theorem that one kernel always has lower effective complexity than another.

Defaults now specify seed 43209 and the zero cutoff on the oriented binary score, including its tie rule. Only the frozen selected model receives a test error. The figure and table compare alternatives using training CV, so the test set does not choose between kernels.

In my Homework 8 review of another student's skill, I found missing YAML, requirements that were omitted from the delivered report, a simplicity tie-break that could favor an extra PCA stage, and overconfident claims from small differences. The new file has valid frontmatter and explicit-only invocation metadata, avoids requiring an artificial two-dimensional boundary plot, defines SE rather than SD, and ends with numerical and visual delivery checks. A compact CV figure is useful with any predictor dimension. The complete skill follows the one-page report.

Explicit invocation

```
Use $stat432-svm at .claude/skills/stat432-svm/SKILL.md.
Analyze homework/week-09/data/saheart.csv; response: chd (0/1).
Predictors: sbp, tobacco, ldl, adiposity, famhist, typea, obesity, alcohol, age.
Encode famhist as Absent=1 and Present=2.
Output directory: homework/week-09/.
Use the skill's default seed, candidate grid, split, folds, classification rule,
selection rule, and reporting requirements.
```

Execution and verification

The skill guided Question 2's 70/30 stratified split, fold-local scaling, 18-setting grid, five-fold training CV, and one-SE rule. Its body contains **43 lines**, including blank lines, after the frontmatter. The one-page report is the next page; it is also saved separately as analysis-report.pdf.

I recomputed mean errors and SEs from saved out-of-fold predictions, checked the split and fold membership, verified scaler means against fitting rows, and confirmed convergence without warnings for all 181 fits (90 per question and one final refit). The selected model is in the one-SE set, and its 36 test mistakes equal 15 false positives plus 21 false negatives. The confusion matrix totals 139 observations. All candidates use exactly the same folds within each question.

Order of analysis. Question 2's defaults and selection were frozen and its test evaluation completed before Question 1's full-data CV. Question 1 subsequently reuses those observations. These analyses are not independent replications, and Question 1 was not used to revise Question 2's final model or report a new test score.

Reproduction and assistance. Run analysis.py, then make_report.py in this folder; fold assignments, predictions, tables, and checks are retained in results/. Python 3.8.6, NumPy 1.24.4, pandas 2.0.3, scikit-learn 1.0.2. Codex assisted with the skill, analysis code, writing, and verification; the skill was AI-assisted rather than independently handwritten. Source: Homework 09 question page and its supplied data package.

SAheart: an SVM selected by the short skill

Setup. 462 rows; 9 predictors; chd classes 302/160; no missing, constant, or duplicate rows. famhist: Absent=1, Present=2. Numeric distances are usable after scaling; independent rows are assumed because no group/time IDs are given. Seed 43209 gives 323 training (211/112) and 139 test (91/48) rows.

Methods. Linear tests a straight boundary, degrees 2-4 add interactions, and radial permits local curvature. Each kernel uses C in {0.1,1,10}; radial crosses gamma in {0.01,0.1} (18 settings). Five shared stratified training folds fit scaling only on fitting rows. Polynomial gamma=1/9, coef0=1; unweighted classes; tolerance 1e-7; predict chd=1 at score >=0. One-SE prefers linear, degrees 2,3,4, then radial, followed by smaller C and gamma.

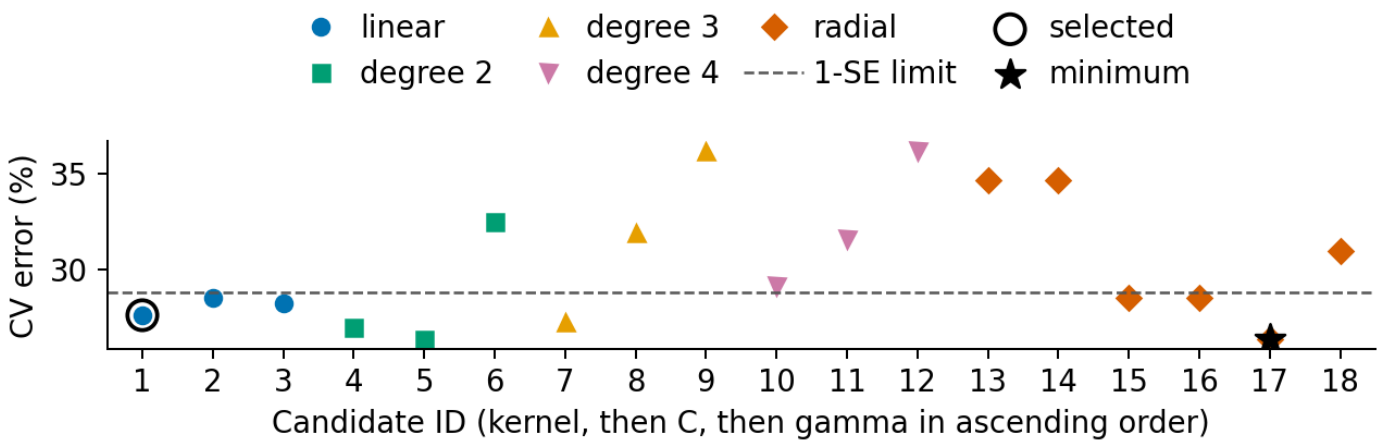


Figure: all 18 training-CV means; ring=selected; star=minimum; dashed=one-SE limit.

ID	Kernel	Settings	CV% / SE pp	Test%
1	Linear *	C=0.1	27.6 / 2.6	25.9
5	Quadratic	C=1	26.3 / 3.4	--
7	Cubic	C=0.1	27.3 / 1.2	--
10	Degree 4	C=0.1	29.1 / 1.7	--
17	Radial	C=10, gamma=0.01	26.3 / 2.5	--

Table: best per kernel; * selected. Selection uses unrounded errors; only the winner is tested.

Selection. Radial C=10, gamma=0.01 minimizes CV error (26.3%, SE 2.5 pp). Its one-SE limit is 28.8%. Linear C=0.1 qualifies (27.6%, SE 2.6 pp) and is selected by the declared simplicity convention. This does not establish that the true boundary is linear.

Held-out results. After refitting on 323 training rows: **36/139 errors = 25.9%**, Wilson 95% interval 19.3-33.8%; 199 support vectors. Confusion matrix (observed rows/predicted columns, order 0,1): **[[76,15], [21,27]]**. Sensitivity 56.2%, specificity 83.5%. Positive cases are harder to detect. The training majority-class error is 34.7%.

Limits. Kernel gaps are small; CV SE=sample SD/sqrt(5) is descriptive because fold fits overlap. Selection CV is optimistic; the held-out interval remains wide. The 34.6% positive fraction and equal-cost cutoff affect sensitivity. One split and 18 settings limit generalization; predictive associations are not causal.

Appendix: complete SKILL.md (1/2)

```
---
name: stat432-svm
description: Compare SVM classifiers and produce a one-page analysis report. Load only
when explicitly invoked with a dataset, response, predictors, coding, and output
directory.
---
# SVM analysis

Invoke only on explicit request. Ask for missing data details; do not infer variable
names.
Keep the analysis and report reproducible; use seed 43209 unless the invocation
overrides it.

1. Audit sample size, class counts, missingness, types, constants, duplicates, and
dependence.
   Judge whether numeric distances and the available class sizes support SVM
classification.
   Explain categorical coding; impute or remove constants inside fitting folds if
needed.
   Use a grouped or temporal split for dependent data; stop if a defensible split is
impossible.
2. Reserve a stratified 30% test set before tuning; use five stratified training folds.
   Reuse identical folds for every candidate. Fit all preprocessing on fitting rows
only.
   Reduce the fold count if class sizes require it and explain the change.
3. Explain kernel choices: linear as a baseline, low-degree polynomials for
interactions,
   and radial for local nonlinear patterns. Use no more than 20 predeclared settings.
   Default grid: C in {0.1, 1, 10} for linear and polynomial degrees {2, 3, 4};
   radial C in {0.1, 1, 10} crossed with gamma in {0.01, 0.1}: 18 settings total.
   For polynomials fix gamma=1/p and coef0=1, with p the encoded predictor count.
   Standardize predictors; use unweighted classes, tolerance 1e-7, and no probability
fitting.
   Any grid adaptation must precede candidate results; state all fixed settings.
4. Default to equal-cost misclassification. For binary classes sorted as (a,b),
   predict b when its oriented decision score is at least zero; otherwise predict a.
   For multiple classes use the library vote rule and report its tie handling.
   State any requested costs or weights before tuning. Scores are not probabilities.
5. Record all fold errors, their arithmetic mean, and sample SD/sqrt(number of folds).
   Call this SE descriptive: overlapping fitting folds prevent an independent-fold CI.
   Let T=min mean CV error + the SE of that minimum-error candidate.
   Among settings with mean error <= T, choose lexicographically by kernel order
   linear, polynomial degree 2, degree 3, degree 4, radial; then smaller C,
   then smaller gamma, then lower CV error. Resolve exact ties by the listed grid order.
   This is an explicit simplicity convention, not a universal complexity ranking.
```

Appendix: complete SKILL.md (2/2)

```
6. Freeze the selection, refit its pipeline on all training rows, then score the test
set once.
   Report only this model's test error, Wilson 95% interval, confusion matrix,
sensitivity,
   specificity, and support-vector count. Do not revise choices using test results.
7. Save analysis-report.pdf and its source in the supplied output directory.
   Use one printed page, 11-point text, and margins of at least 1.5 cm.
   Include setup, kernel rationale, grid, folds, selection rule, and the selected
settings.
   Include one compact figure of all CV errors and T; mark the selected and minimum
rows.
   Include one table of each kernel's best CV candidate plus the selected row if
different.
   Give CV mean and SE, selected test results, a training majority baseline, and
limitations.
   Interpret differences cautiously; avoid causal claims or unsupported superiority
claims.
8. Check counts, folds, leakage, selection, arithmetic, convergence, and all report
requirements.
   Keep the code and full CV table. Render and visually inspect the PDF; shorten prose
   or reduce the figure before changing font size. Never silently omit required
evidence.
```